

Dimensions of Quantum codes from Cayley graphs over $\mathbb{F}_2^r$

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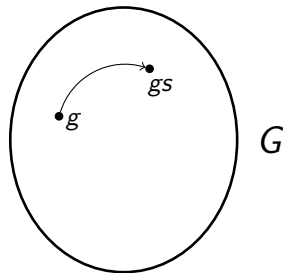
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Cayley graphs

Definition

Let G be a group and S a subset of G . The **Cayley graph** $\text{Cay}(G, S)$ is the graph whose vertices are the elements of G , and where two vertices g, h are connected by an edge if and only if there exists an element $s \in S$ such that $h = gs$.



Fix $G = \mathbb{F}_2^r$ abelian.

- Every element is an involution $\rightarrow$ the graph is undirected.
- The adjacency matrix A satisfies $A^2 = 0 \iff |S|$ even.

The homology of a Cayley graph

Definition (Quantum CSS code)

A **quantum CSS code** $\mathcal{C} = (\mathcal{C}_X, \mathcal{C}_Z)$ is defined by two matrices H_X and H_Z such that $\text{rowspan}(H_X) \perp \text{rowspan}(H_Z)$.

$$H_X = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}, \quad H_Z = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}.$$

The two codes $\mathcal{C}_X$ and $\mathcal{C}_Z$ are $\ker H_X / \text{Im } H_Z^T$ and $\ker H_Z / \text{Im } H_X^T$, respectively.

Alternatively, we can define $\mathcal{C}$ as the (co)homology of the chain complex:

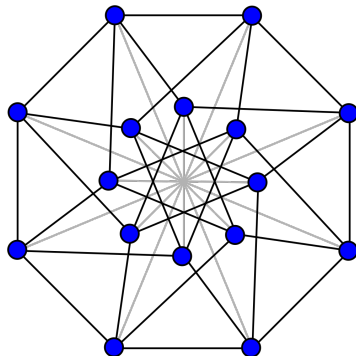
$$\mathfrak{C} : \mathbb{F}_2^{n_X} \xrightarrow{H_Z^T} \mathbb{F}_2^n \xrightarrow{H_X} \mathbb{F}_2^{n_Z}, \quad \text{here: } H_X = H_Z = A.$$

$A : \mathbb{F}_2^{2^r} \rightarrow \mathbb{F}_2^{2^r}$, such that $A^2 = 0 \rightarrow$ A CSS code $\mathcal{C}$.

Parameters $[[n, k, d]]$ where $n = |G| = 2^r$, $k = \dim \ker A / \text{Im } A$ and d is the minimum weight of a non-trivial homology class.

History

- MacKay et al. (2007): $\text{Cay}(\mathbb{F}_2^r, S)$, where S is the set of columns of a parity-check matrix H of a classical code
→ quantum LDPC codes.
- Couvreur, Delfosse, and Zémor (2013): studied the parameters of $\mathcal{C}$ in the case of repetition code $[r, 1, r]$ and obtained $[[2^r, 2^{\frac{r}{2}}, 2^{\frac{r}{2}}]]$.



Barg, Zémor (2022) give necessary conditions for high rate quantum codes:

Theorem

Let $\text{Cay}(\mathbb{F}_2^r, S)$ be a Cayley graph over $\mathbb{F}_2^r$ with set of generators S . Let $\mathcal{C}$ be the quantum code obtained as homology of the adjacency matrix $A(\Gamma)$. Then

- ① $R(\mathcal{C}) > 0 \implies |S|$ and all row-weight of H are even; (Lowzow)
- ② $R(\mathcal{C}) > (1 - 2^{1-k}), k = 2, 3, \dots \implies (\mathcal{C}^\perp)^{*(k-1)} \subset \mathcal{C}$.

What about the converse?

Contribution:

- ① is an equivalence;
- ② is not a sufficient condition.

Group algebras

Given A the adjacency matrix of a Cayley graph:

$$A \in \mathbb{F}_2^{2^r} \cong \{f : \mathbb{F}_2^{2^r} \rightarrow \mathbb{F}_2\} \cong \mathbb{F}_2[\mathbb{F}_2^{2^r}] \cong \mathbb{F}_2[y_1, \dots, y_r] / \langle y_1^2 - 1, \dots, y_r^2 - 1 \rangle$$

$\implies A := f$ is an element in the quotient ring.

Example

The construction coming from the repetition code of length $n = 2^r$ corresponds to the polynomial

$$f = 1 + \sum_i y_i + \prod_i y_i,$$

if we apply $x_i = y_i + 1$:

$$f = \sum_{\deg m \geq 2} m, \quad m \text{ monomial.}$$

→ Apply

$$x_i = y_i + 1$$

and study the homology $H(f)$ where $f \in \mathbb{F}_2[x_1, \dots, x_r]/\langle x_i^2 \rangle$.

Definition

The **valuation** $v(f)$ of a polynomial f is the minimum degree of its monomials.

Reformulation of the theorem

$\mathcal{C}$ is the homology of f , then

- ① $R(\mathcal{C}) > 0 \iff v(f) \geq 2;$
- ② $R(\mathcal{C}) > (1 - 2^{1-k}), k = 2, 3, \dots \implies v(f) \geq k.$

Characterization of non-trivial quantum codes over $\mathbb{F}_2^r$

Theorem (T., Zémor)

Let $f \in \mathbb{F}_2[x_1, \dots, x_r]/(x_1^2, \dots, x_r^2)$ and $\mathcal{C}$ the quantum code obtained as homology of the adjacency matrix A defined by f . Then

$$R(\mathcal{C}) > 0 \iff v(f) \geq 2.$$

Example

- If $f = x_1$, then $\ker f = \text{Im } f$,
- If $f = x_1 x_2$, then $\ker f$ is generated by x_1 and x_2 , while $\text{Im } f$ is generated by $x_1 x_2$. Hence $\dim(H(f)) = 3 - 1 = 2$;
- If $f = x_1 + g$ also $\ker f = \text{Im } f$,
- If $f = x_1 x_2 + x_3 g$ then, $0 \neq x_1 x_3 \in H(f)$.

We showed that $\dim H(f) \geq \dim H(g)$ when $f = x_r h + g$.

Some simple cases

Proposition

Let $f = g + h$, $v(g), v(h) \geq 1$ and g, h defined on disjoint sets of variables. Then

$$\dim(H(f)) = \dim(H(g)) \cdot \dim(H(h)).$$

For instance, if $f = x_1x_2 + x_3x_4x_5$, then

$$H(f) = H(x_1x_2) \otimes H(x_3x_4x_5) = \text{Span}(\{x_1, x_2\} \times \{x_3, x_4, x_5, x_3x_4, x_4x_5, x_3x_5\}).$$

In particular, the second condition is too weak.

Example

The polynomial $f = x_1x_2x_3 + x_4x_5x_6 + x_7x_8x_9$ defines the quantum code $H(f)$ of rate

$$R(H(f)) = \frac{\dim(H(f))}{2^9} = \left(\frac{2^3 - 2}{8}\right)^3 = \left(\frac{3}{4}\right)^3 < \frac{1}{2}.$$

When things are tractable: the quadratic case

Quadratic forms over finite fields are well-classified.

In the square-free case there is only one non-degenerate form.

The **tensor rank** of a quadratic form is the minimal number of monomials of degree 2 needed to write it, and it is at most $\lfloor r/2 \rfloor$.

Corollary (T., Zémor)

A quadratic form of algebraic (tensor) rank $k \leq r/2$ has homology of dimension 2^{r-k} .

More precisely, the homology of such quadratic forms is explicitly generated by the monomials:

$$S = \{x_{i_1}x_{i_2} \cdots x_{i_k}, x_{i_j} \text{ divides } m_j, 1 \neq x_{i_j} \neq m_j, \deg m_j = 2\} \otimes R,$$

where R is the remaining space.

Many polynomials are quadratic forms

There exists a *quasi*-linear isomorphism that reduces many other polynomials to quadratic forms (including the repetition code).

Proposition

Let $f = q + \tilde{f}$ be a square-free polynomial such that $v(\tilde{f}) \geq 3$, q a full rank quadratic form, and $\tilde{f}$ does not involve any variable not appearing in q . Then we can write $f = f_1 f_2 + \cdots + f_{2k-1} f_{2k}$ such that $f_i = x_i(1 + h_i) + g_i$ with $v(g_i) \geq 2$ and $v(h_i) \geq 1$ and both not involving the variables x_j for $j \leq i$.

Corollary

The dimension of the homology of any polynomial that contains a full rank quadratic form is 2^{r-k} .

And cubic forms?

There exists a lower bound in terms of series of binomial series for the dimension of the quantum code defined by homogeneous polynomials of degree k .

- $k = 2$ the bound is tight and depends on the algebraic tensor rank.
- $k = 3$:

$$\begin{cases} R(C) \geq 3^{\frac{r-1}{2}} & r \text{ odd} \\ R(C) \geq 2 \cdot 3^{\frac{r}{2}-1} & r \text{ even.} \end{cases}$$

Conjecture

The bound is close to optimal, but not tight for all values of r (e.g. $r = 7, 9$).

Unsolved problems

- For quartic forms, we obtain the following bound:

$$R(\mathcal{C}) \geq \zeta^r$$

Where ζ is the largest eigenvalue of the matrix

$$\begin{bmatrix} 0 & 4 & 6 & 4 \\ -4 & 0 & 4 & 6 \\ -6 & -4 & 0 & 4 \\ -4 & -6 & -4 & 0 \end{bmatrix}.$$

What about higher degree forms?

- The **minimum distance** of such quantum codes.
For quadratic forms of maximal algebraic rank it equals $2^{\lfloor r/2 \rfloor}$.
Is it the same for all polynomials containing a full rank quadratic form? What about higher dimensional forms?
- How does the dimension behave for inhomogeneous polynomials?
- Dimension Rank $\leftrightarrow$ Tensor Rank for cubic forms?